



MOSAIC The Garden of Forking Tissues

A specification-curve multiverse reveals — and repairs — the fragility of biological conclusions in spatial transcriptomics

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THE PROBLEM

• Every spatial-transcriptomics analysis makes ~12 individually defensible choices — normalization, gene selection, dimensionality, neighborhood graph, clustering algorithm & resolution, and spatially-variable-gene (SVG) test. This is the “garden of forking paths”: the same data can yield different biological conclusions depending on which path is taken. Researchers walk ONE path and report its endpoint as the answer — the analytical uncertainty is never reported.

HYPOTHESES

• [object Object],[object Object][object Object],[object Object][object Object],[object Object]

THE SPATIAL STABILITY SCORE (S³)

• Per spot: the fraction of pipelines that agree on its domain, after aligning arbitrary cluster labels across pipelines by optimal (Hungarian) assignment.[object Object],[object Object],[object Object],[object Object]

$$S^3(i) = (1/N) \cdot \max_{\ell} | \{ s : \pi_{\ell}(c_s(i)) = \ell \} |$$

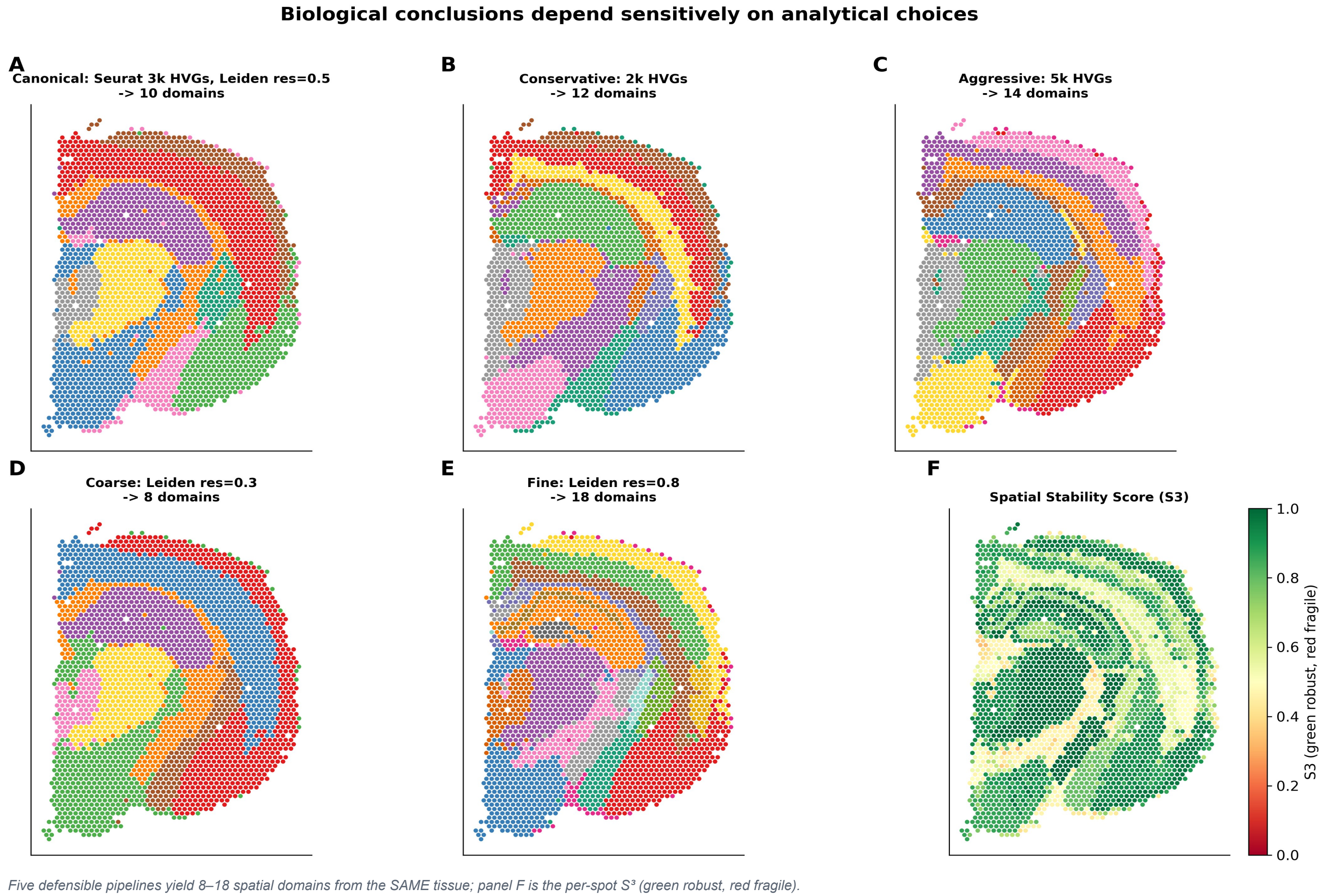
METHODS

• 30 curated, literature-defensible pipelines × 4 tissues (2 mouse-brain sections, human breast cancer, glioblastoma). Every specification seeded & bit-reproducible; no leakage — each re-derives from raw counts. A null for every claim: label-permutation, bootstrap CIs, and a depth-stratified control. Validated against reference anatomy AND independent board-certified pathology.

30
defensible pipelines

4
tissues / cancers

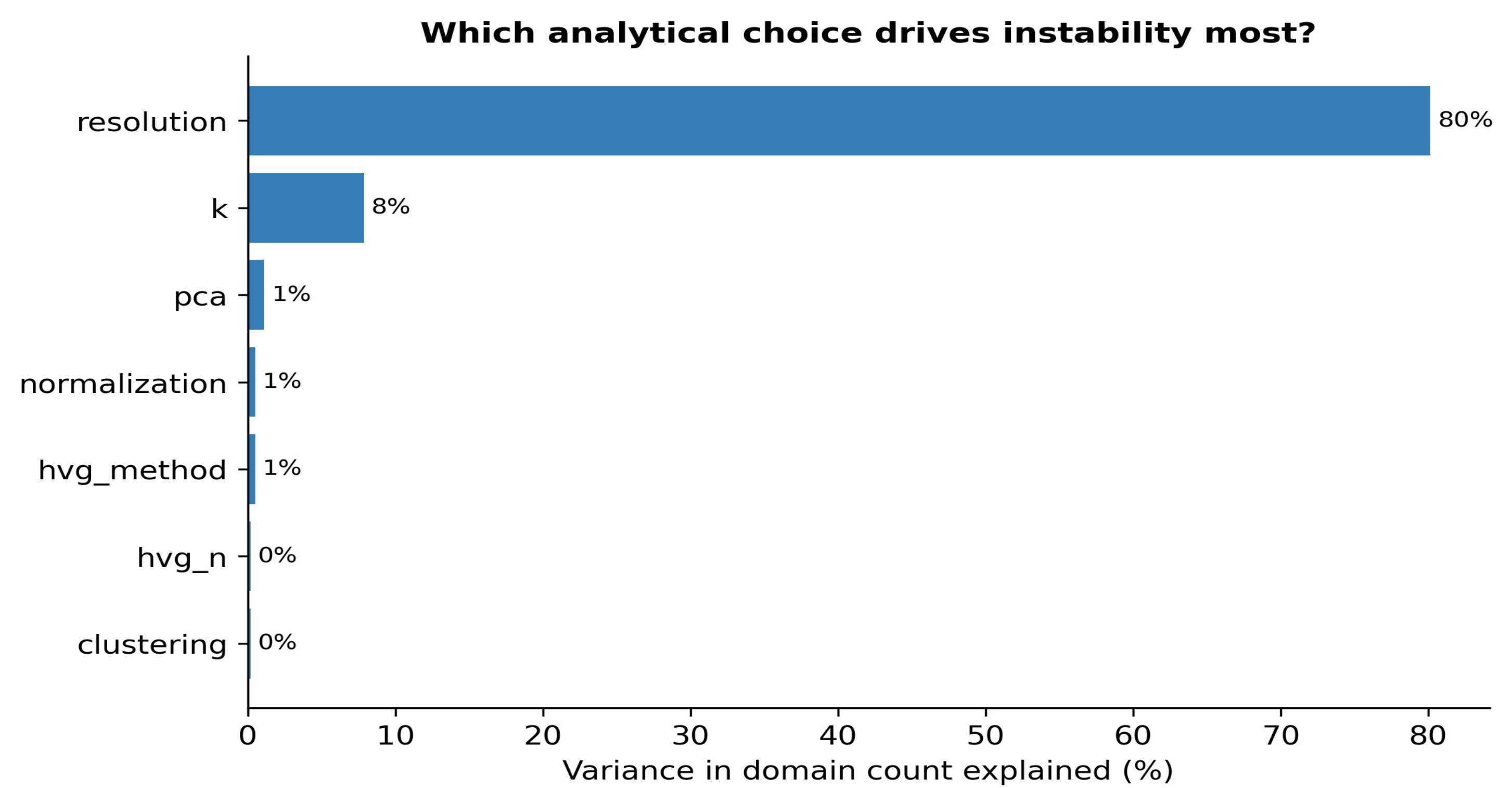
ONE TISSUE — MANY ANSWERS



INSTABILITY IS LARGE — AND IT'S BIOLOGY

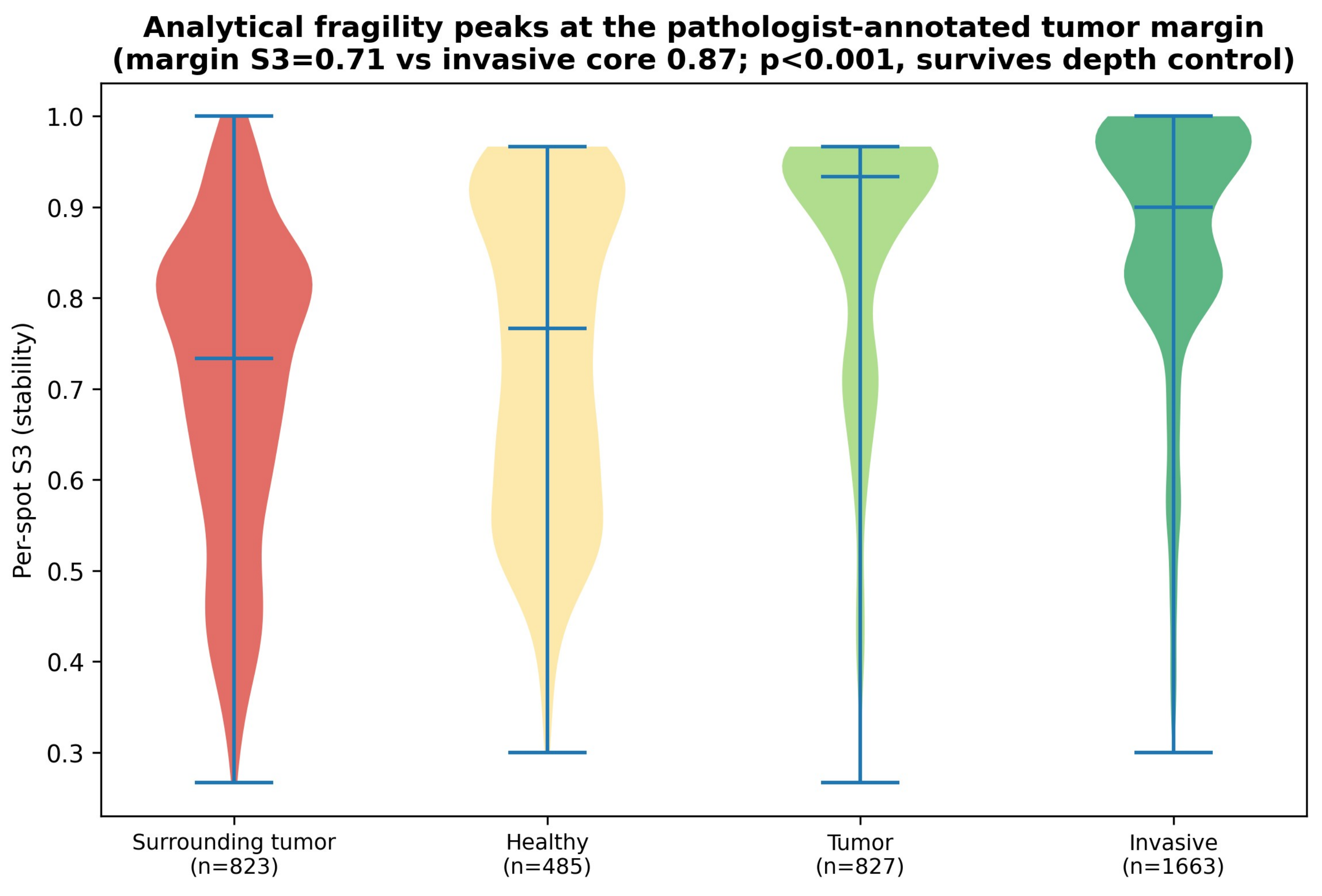
• Recovered domains range 8–23 (mean 13.7); mean pairwise Adjusted Rand Index between pipelines is only 0.68. Only ~60% of spots receive a robust domain across the multiverse.[object Object],[object Object]

ONE CHOICE DOMINATES (H2)



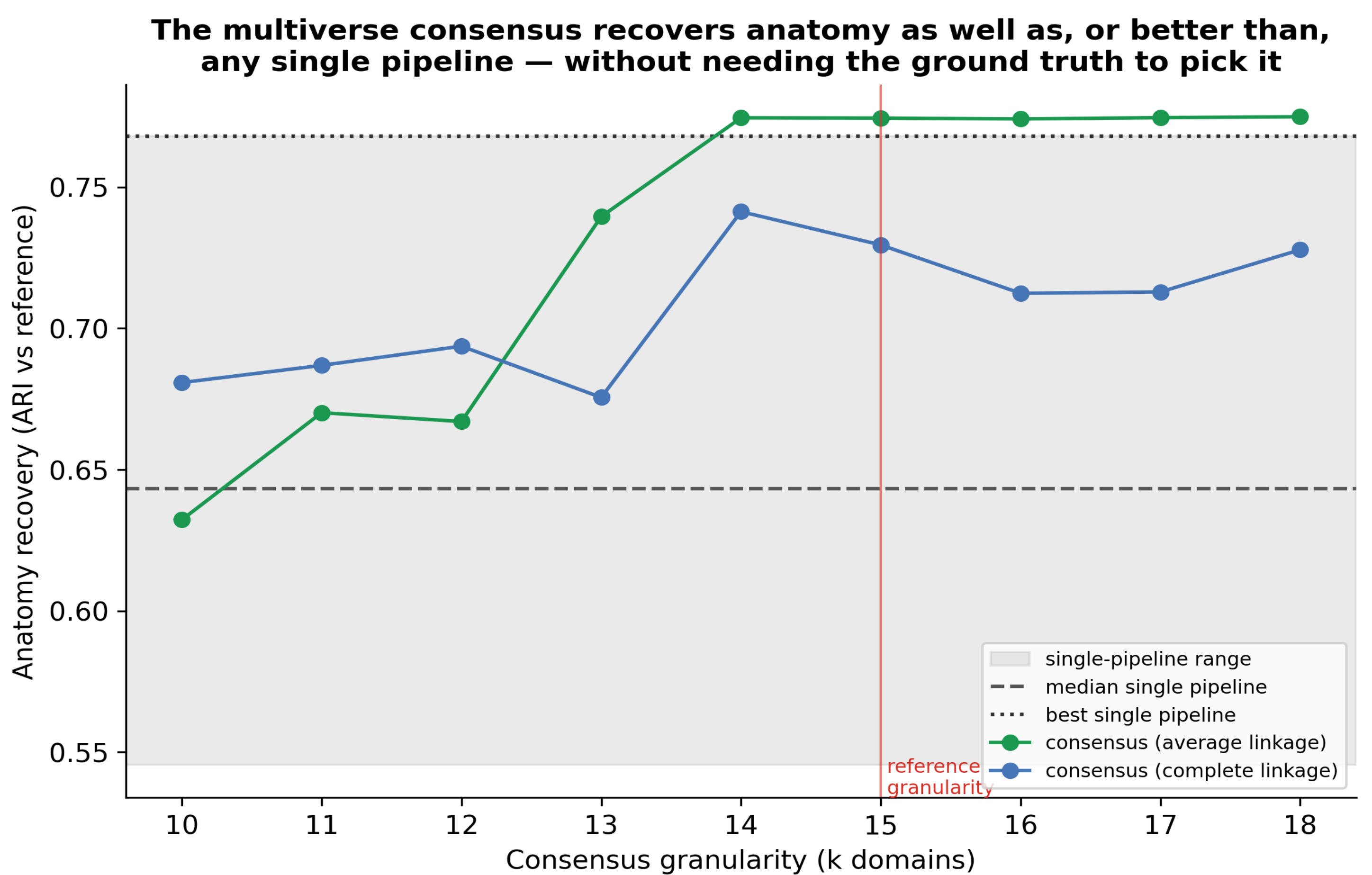
• [object Object],[object Object][object Object],[object Object]

CLINICAL VALIDATION (H1)



• [object Object],[object Object][object Object],[object Object]The place analysis is least trustworthy is exactly where clinicians care most.

THE BREAKTHROUGH — A BETTER ESTIMATOR



• Rather than pick one pipeline, take the multiverse CONSENSUS (co-association clustering).[object Object]S³ predicts where the consensus errs (AUC 0.72): uncertainty-quantified for free.

CONCLUSIONS

• Spatial-transcriptomics conclusions are contingent on analytical choices — most at clinically critical margins.[object Object],[object Object][object Object],[object Object]